Analysis

February 21, 2025

Malaysia: from sceptic to believer

How Malaysia is advancing its role in Southeast Asia's interconnected future

Energy Policy

System Modelling

Summary

Malaysia has started exporting renewable electricity to Singapore, signalling a major shift in its energy strategy and a key step forward for regional grid integration in Southeast Asia.

With stronger momentum in favour of the ASEAN Power Grid, power system modelling tools like TZ-APG can help stakeholders in Malaysia examine the merits of different interconnection options and identify the most cost-effective and climate-friendly candidates to prioritize development.

The TZ-APG v1 model highlights further potential in the Peninsular Malaysia-Singapore interconnection, and the possible benefits to Malaysia if the multilateral cross-border power trading scheme between Brunei Darussalam, Indonesia, Malaysia, and the Philippines becomes a reality. Modelled outputs also help to assess carbon leakage risks of regional grid integration.

Willing buyer, willing seller

It is no exaggeration to say that the ASEAN Power Grid (APG) discourse has been revived by a power couple, Singapore and Malaysia. Singapore has emerged as a clean energy-hungry consumer seeking long-term electricity imports, while neighbouring Malaysia is endowed with the geographical and resource advantages needed to become a key player in the region's cross-border trading landscape.

In 2024, a significant milestone in the trading relationship between the two countries was reached with Malaysia officially commencing electricity exports to Singapore. Under the regional multilateral trading scheme Lao PDR-Thailand-Malaysia-Singapore Power Integration Project (LTMS-PIP) Phase 2, Malaysia is now directly contributing 100MW of export capacity to the city-state in addition to the electricity sourced from Lao PDR. Separately, following a bilateral agreement between Tenaga Nasional Berhad (TNB) and Sembcorp Power, Malaysia started delivering up to 50MW of renewable energy to Singapore from December 2024. Both arrangements, which rely on the existing grid infrastructure, will remain in effect for an initial period of two years.

This is an impressive turnaround from the dynamic seen four years ago when Malaysia banned clean power exports to Singapore. The shift was formalised with the launch of Malaysia's National Energy Transition Roadmap (NETR) in 2023, which stated that Malaysia could promote the development of renewable energy domestically by becoming a regional hub for cross-border renewable energy trading.

Ever since, the government has been backing the NETR strategy with key resources. The Energy Exchange Malaysia (ENEGEM) was established in late 2023, serving as the centralised platform for cross-border green electricity sales from the Peninsular grid to neighbouring countries. Just a year later, in December 2024, it registered the maiden renewable energy export to Singapore. This marks an encouraging start that could effectively advance the region's APG agenda.

Sleeping giant

Malaysia is well-positioned to be a clean energy trading hub. Straddling both the Indochinese Peninsula and the island of Borneo, Malaysia shares land and maritime borders with six Southeast Asian countries. However, grid interconnection and cross-border electricity trade were not a strategic power system development priority until recently.

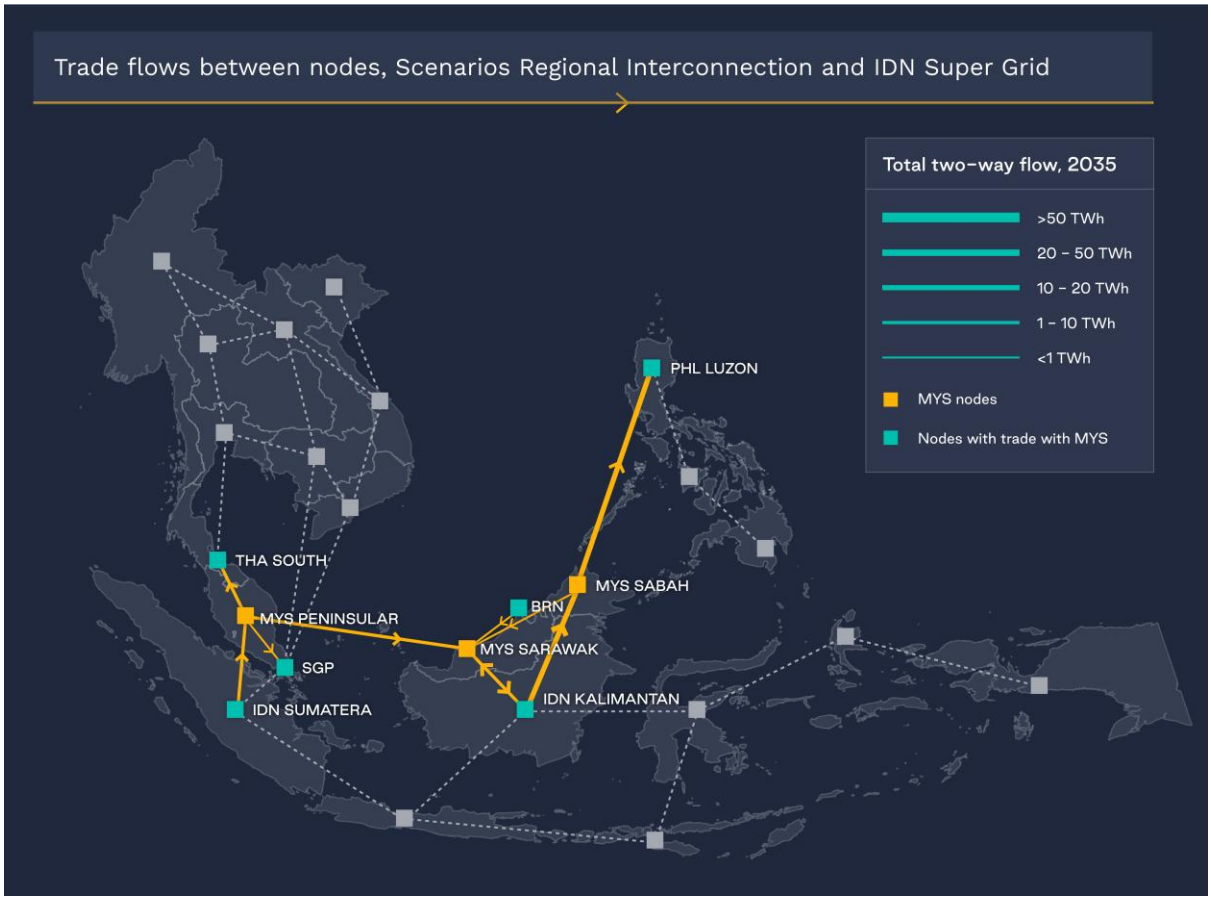
Before 2023, cross-border flows were only registered with two neighbours: mostly exports to Indonesia's West Kalimantan region, via the Sarawak grid, and limited two-way trade with Thailand via the Peninsular Malaysia grid. Annual export volumes have been modest, averaging less than one terawatt hour (TWh) per year, equivalent to just 2.5% of total regional trade. This highlights the system balancing role of the trade, which the government is now seeking to move beyond.

Another distinct feature is that Malaysia's cross-border trade with its neighbours has typically followed the grid-to-grid integration model, in which the two power systems would connect with each other. This is different from the independent power producer IPP-to-grid scheme that has been dominating the transactions in northern Southeast Asia of late. Under an IPP-to-grid scheme, a power generator would be built exclusively for export purposes and would only feed into the national grid of the importing country. In contrast, Malaysia's approach actually supports the realisation of the APG concept of regional grid interconnectivity at the highest level.

Assessing potential with TZ-APG v1

Malaysia could play a much larger role in Southeast Asia's electricity trading landscape than it does today given the resources at hand and the government's clear policy pivot.

The ASEAN Interconnection Masterplan Study III (AIMS III) has identified a shortlist of new interconnectors worth pursuing by Malaysia in the realisation of the APG. This includes grid linkages between MYS Peninsular and MYS Sarawak; MYS Peninsular and IDN Sumatera; MYS Sabah and IDN Kalimantan; and MYS Sarawak and Brunei. If implemented, this grid network could propel Malaysia into a regional 'superconnector'. Unfortunately, due to insufficient political interest, none of these projects have advanced beyond the research stage.



TZ-APG v1 illustrates Malaysia's potential as a key regional hub and superconnector

Open

Models like TZ-APG can help stakeholders examine the merits of these interconnection candidates. For full details on TZ-APG v1 model design and constraints, see **here.**

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In the first iteration, TZ-APG v1, the modelled outputs offered the following insights:

Malaysia should optimise its existing grid connection with Singapore

TZ-APG v1 highlighted that MYS Peninsular and Singapore's existing 1GW transmission line is currently severely under-utilised. In our business-as-usual (BAU) scenario,

Malaysia's exports could supply around 7% of Singapore's total demand in the modelled year 2035. In addition, the model suggests that the optimal transmission capacity would be 2GW, facilitating a doubling in the export volume per the Enhanced BAU' scenario. Singapore has announced ambitious plans to build subsea cables to connect its power system with other countries; however, until these plans become reality, Malaysia is uniquely positioned to serve Singapore's growing electricity needs.

Malaysia is vital to regional multilateral trading schemes

To date, Malaysia has been indispensable to the LTMS-PIP scheme, functioning as the intermediary node transmitting electricity from Lao PDR to Singapore. Future initiatives such as Southeast Asia's second multilateral cross-border power trading scheme, the Brunei Darussalam–Indonesia–Malaysia–Philippines Power Integration Project (BIMP-PIP), will also depend on Malaysia's involvement to succeed.

In the 'Regional Interconnection' grid scenario of TZ-APG v1, the results spotlight the IDN Kalimantan–MYS Sabah–PHL Luzon transmission corridor which, if realised, could bring about significant decarbonisation impact for the region. IDN Kalimantan could capitalise on its hydropower resources and export to PHL Luzon, replacing higher-cost gas power generation in the Philippines. Unlocking this corridor would need Malaysia's support as the "wheeling" partner. While technically and financially ambitious, given the need for subsea interconnection between the Borneo and Luzon islands, the transmission volume could be significant — 20TWh per year — which is more than Malaysia's trade with any other partner.

Peninsular and Sarawak grid linkage needs more consideration

Under the current design and configuration, the TZ-APG v1 model indicates that the submarine cable connecting Malaysia's two grid systems would yield only modest benefits. This is likely because each grid region has sufficient resources to service local demand. The simulation shows that trading volumes would be insignificant, with only one-way flow registered from MYS Peninsular, suggesting a weak economic case for the project.

Carbon leakage risk is real

Grid-to-grid electricity trade carries the risk of carbon leakage, as it becomes harder to verify the source of supply compared to the isolated IPP-to-grid interconnection approach. Theoretically, demand for exports could boost the operations of coal and gas power plants, resulting in higher local emissions if the grid's power mix does not

decarbonise more quickly.

This is a valid concern in the case of MYS Peninsular where coal and gas power will continue to dominate the grid, and the reserve margin is expected to remain high (projected at 28-36% until 2030). So far, Singapore seeks to address this with the mandated use of Renewable Energy Certificates (RECs) as evidence that the energy is sustainably sourced from MYS Peninsular. The two countries are also in talks to design a framework for the mutual recognition of green attributes associated with cross-border electricity trade. This would be a pathfinder for the region, and might lend confidence to other member states to pursue grid-to-grid integration as they gradually decarbonise the power mix.

In 2025, TransitionZero will be working on updates to TZ-APG v1, making sure that our model incorporates the latest data and market dynamics. Watch this space for the new releases and don't hesitate to reach out if you have any suggestions for TZ-APG v2.